Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/03/2005
PI Name:	Kelliher, Warren C. M.S.		
Project Title:	Development of an Inflatable Multifunctional Damage Tolerant Wall Structure with High Radiation Attenuation Properties		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Langley Research Center		
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City:	Hampton	**State:**	VA
Zip Code:	23681-2199	**Congressional District:**	3
Comments:	Deceased, Nov 2010 (rje 9/2/16).		
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-UNSOL
Start Date:	07/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Barghouty, N	**Contact Phone:**	256 544-0238
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Thibeault, Sheila Ann (NASA Langley Research Center) Carlberg, Ingrid A. (NASA Langley Research Center)		
Grant/Contract No.:	Intramural		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This task aims to characterize candidate multifunctional materials and structural configurations for their suitability to be inflatable, provide damage tolerance, and provide adequate radiation shielding against galactic heavy ion radiation. Sandwich inflatable construction is considered in this study. It provides a versatile structural approach in which spectra fabric facesheets are joined to an interior polyethylene core to assure cabin atmospheric integrity. The mechanical demands on the core are low and should permit the core to be composed of optimum shielding material foam. The mechanical and shielding performance may be tailored to meet different service requirements by changing the thickness of structural and inner and outer layers. The goal of this effort will be to approach a near pure polyethylene composition with known good radiation attenuation properties. In this task, we shall address the multifunctional optimization of inflatable sandwich composites for combined structural and radiation shielding applications. Mechanical testing and analysis will be used to characterize the suitability of the configurations under structural requirements, while radiation transmission testing with protons and HZE ion beams will be used to characterize shielding performance. The primary accelerator facility to be used will be the NASA Space Radiation Laboratory at the Brookhaven National Laboratory. The primary significance of this work is the generation of materials property data, including radiation shielding effectiveness, of not only the multilayered wall but also the individual constituent layers. The resulting materials property database will provide a design tool for designers of in-space habitats and other space structures requiring radiation shielding.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The primary application of this technology is for in-space habitats. These habitats could be used for Earth observation and monitoring.
Task Progress:	Task Milestones and Deliverables: Year 1 (FY04): Milestones: (1) Search for candidate materials: Flexible, open cell, polymeric foams. (Completed) (2) Select suitable foams based on radiation protection potential and availability. (Completed) (3) Acquire from manufacturer suitable foam samples for radiation testing. (Completed) (4) Test foams with radiation beams at the NASA Space Radiation Laboratory. (Completed) (5) Perform radiation transmission calculations for comparison with experimental data. (Initiated) Deliverables: (1) Power Point presentation on results to Program Manager at Fiscal Year end. (Completed at August PI conference) Year 2 (FY05): Milestones: (1) Select and acquire additional foams for testing based on radiation protection potential and additional properties, such as flame retardation and lack of moisture affinity. (Completed) (2) Select and acquire adhesives based on radiation protection potential and capability to bond to polymer foams. (Initiated) (3) Test foams for flame retardation. (Initiated) (4) Complete literature search to identify properties of Transhab. (Completed) Deliverables: (1) Task Book submission. (Completed)
Bibliography Type:	Description: (Last Updated: 02/28/2011)
Papers from Meeting Proceedings	Wilson JW, Glaessgen EH, Jensen B, Orwoll R, Saether E, Adams DO, Humes DH, Kelliher WC, Thibeault SA, Anderson BM, Nealy JE, Tripathi RK. "Next Generation Shielding Materials for Near-Earth Infrastructure." AIAA Space 2003 Conference and Exposition, Long Beach, California, Sep. 23-25, 2003. 2003 AIAA Meeting Papers, AIAA Space 2003 Conference, Vol. 8, No. 4, AIAA paper AIAA-2003-6258, September 2003. , Jan-2004
Papers from Meeting Proceedings	Thibeault SA, Hou T-H, Wilson JW, Jensen BJ, Kiefer RL, Chang CK, Hunter A. "Hot-Press Fabrication of Polymer Fiber/Matrix Composites for Multifunctional Radiation Shielding." Space 2004 Conference and Exhibit, San Diego, California, Sep. 28-30, 2004. Space 2004 Conference and Exhibit, Vol. 9, No. 20. AIAA paper AIAA-2004-6028. September 2004. , Sep-2004